## ✅ What is MySQL Table Partitioning?

**Partitioning** means:

Splitting one large table into smaller pieces called *partitions*.

Each partition:

- Stores part of the data
- Has its own indexes
- Works like a mini-table

👉 But logically, it is still **one single table**.

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## 🎯 Why Use Partitioning?

Partitioning is useful when:

- Table is **very large**
- Table is **growing fast**
- Queries scan large amounts of data
- You manage historical data (e.g., yearly orders)

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## 🔥 Main Benefits

✓ Faster queries (scans only needed partitions)
✓ Smaller indexes → better performance
✓ Easy deletion of old data
✓ Easier maintenance
✓ Better storage management

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## 🧠 Simple Example Idea

Imagine a huge cupboard full of files.

Without partitioning:
👉 All files in one big drawer.

With partitioning:
👉 Files separated by year (2019, 2020, 2021…).

Now if you need 2021 files, you open only that drawer!

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## 📌 Types of Partitioning in MySQL

MySQL supports:

1️⃣ RANGE
2️⃣ LIST
3️⃣ HASH
4️⃣ KEY

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## 1️⃣ RANGE Partitioning

🔷 **Divides data based on a range of values**

Usually used for:

- Dates
- Years
- Numeric ranges

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**Example:**

```
PARTITION BY RANGE (YEAR(order_date)) (
  PARTITION p0 VALUES LESS THAN (2019),
  PARTITION p1 VALUES LESS THAN (2020),
  PARTITION p2 VALUES LESS THAN (2021),
  PARTITION p3 VALUES LESS THAN (2022)
);
```

👉 Orders are stored by year.

Example:

- 2018 → p0
- 2019 → p1
- 2020 → p2

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## 2️⃣ LIST Partitioning

🔷 **Divides data based on specific values**

Used when:

- Categories exist
- Departments

- Regions

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**Example:**

```
PARTITION BY LIST COLUMNS(department) (
   PARTITION p0 VALUES IN ('HR', 'Finance'),
   PARTITION p1 VALUES IN ('Engineering', 'Sales'),
   PARTITION p2 VALUES IN ('Marketing', 'Support')
);
```

👉 Employees are grouped by department.

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### 3️⃣ HASH Partitioning

🔷 **Distributes rows evenly using a hash function**

Good for:

- Even data distribution
- Load balancing

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**Example:**

```
PARTITION BY HASH(user_id) PARTITIONS 4;
```

👉 MySQL automatically spreads rows into 4 partitions.

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### 4️⃣ KEY Partitioning

Similar to HASH, but:

👉 MySQL uses its own internal hashing function.

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**Example:**

```
PARTITION BY KEY(product_id) PARTITIONS 4;
```

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### 🛠 How to Implement Partitioning

**Step 1: Choose Partition Key**

Select column:

- Frequently used in WHERE
- Used in filtering (e.g., order_date)

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**Step 2: Create Table with PARTITION BY**

Use:

CREATE TABLE ... PARTITION BY ...

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**Step 3: Define Partition Rules**

Specify:

- Type (RANGE, LIST, HASH, KEY)
- Boundaries or number of partitions

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🔄 **Managing Partitions**

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1️⃣ **Add Partition**

```
ALTER TABLE orders
ADD PARTITION (
 PARTITION p4 VALUES LESS THAN (2023)
);
```

👉 Add new year partition.

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2️⃣ **Drop Partition**

```
ALTER TABLE orders DROP PARTITION p0;
```

👉 Deletes entire partition instantly.

⚠️ Very useful for deleting old data quickly.

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3️⃣ **Merge Partitions**

```
ALTER TABLE orders
REORGANIZE PARTITION p1, p2 INTO (
PARTITION p1_2 VALUES LESS THAN (2021)
```

);

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### 4️⃣ Split Partition

ALTER TABLE orders

REORGANIZE PARTITION p3 INTO (

PARTITION p3_1 VALUES LESS THAN (2022),

PARTITION p3_2 VALUES LESS THAN (2023)

);

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## 🚀 Major Benefits Explained Simply

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### 🔷 Faster Queries

If query is:

WHERE order_date = '2021-05-10'

MySQL scans only 2021 partition
👉 Not entire table

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### 🔷 Easier Data Management

Want to delete 2018 data?

Instead of:

DELETE FROM orders WHERE YEAR(order_date) = 2018;

Just:

ALTER TABLE orders DROP PARTITION p0;

👉 Much faster

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### 🔷 Better Availability

You can:

- Optimize one partition
- Repair one partition
- Backup one partition

Without affecting others.

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### 🔷 Efficient Indexing

Each partition has:

- Smaller index
- Faster search
- Less disk usage

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### ⚠️ Important Rules & Limitations

- Partition key must be part of PRIMARY KEY
- Not all storage engines support partitioning
- Too many partitions can slow performance
- Use partitioning only for large tables

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### 📊 When NOT to Use Partitioning

❌ Small tables
❌ Tables under few million rows
❌ When no filtering column exists

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### 🧠 Easy Comparison

| Without Partitioning | With Partitioning |
|---|---|
| One big table | Multiple smaller pieces |
| Full table scan | Partition scan |
| Slow for large data | Faster for filtered queries |

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### 🎯 Quick Summary

Partitioning = Breaking large table into smaller logical parts

Types:

- RANGE → by range
- LIST → by category
- HASH → even distribution
- KEY → MySQL internal hash

Best for:

- Large datasets
- Date-based filtering
- Archiving old data